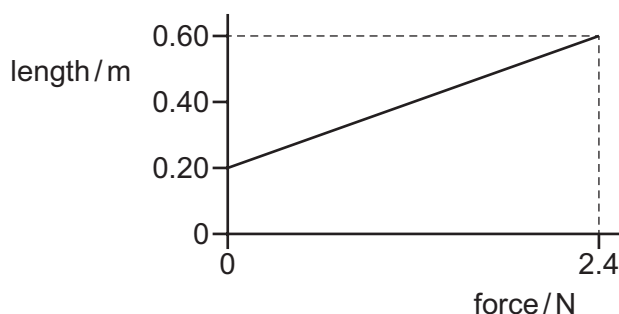


- 20 A parachutist is falling at constant (terminal) velocity.

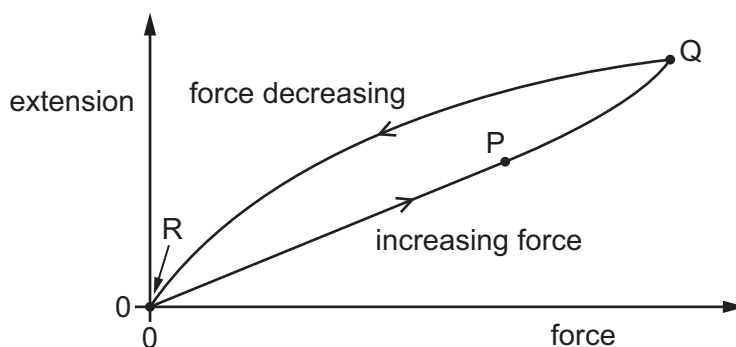
Which statement is **not** correct?

- A Gravitational potential energy is converted into kinetic energy of the air.
 - B Gravitational potential energy is converted into kinetic energy of the parachutist.
 - C Gravitational potential energy is converted into thermal energy of the air.
 - D Gravitational potential energy is converted into thermal energy of the parachutist.
- 21 A student investigates a spring. The variation of the length of the spring with the force applied to the spring is shown.



What is the spring constant of the spring?

- A 0.17 Nm^{-1}
 - B 0.25 Nm^{-1}
 - C 4.0 Nm^{-1}
 - D 6.0 Nm^{-1}
- 22 The graph shows the variation with force of the extension of a wire.



The force is gradually increased to a maximum at Q and then gradually decreased to zero at R.

Which statement is correct?

- A Along the line PQ, the wire obeys Hooke's law.
- B Along the line RP, the spring constant is equal to $\frac{\text{extension}}{\text{force}}$.
- C The wire has elastic deformation at point Q.
- D The work done in stretching the wire to P is equal to (force $\times$ extension) at point P.